Introduction to the Tidyverse

Import, wrangle, model, and communicate data

2026-07-30

##


dplyr
ggplot2
TIBBLE
readr
tidyr
purrr

Working with data in R

the tidyverse is a collection of *friendly and consistent* tools for data analysis and visualization.

They live as R packages, each of which does one thing well.

`library(tidyverse)` will load
the core packages:

ggplot2, for data visualisation.

dplyr, for data manipulation.

tidyr, for data tidying.

readr, for data import.

purrr, for functional programming.

tibble, for tibbles, a modern re-imagining of data frames.

stringr, for strings.

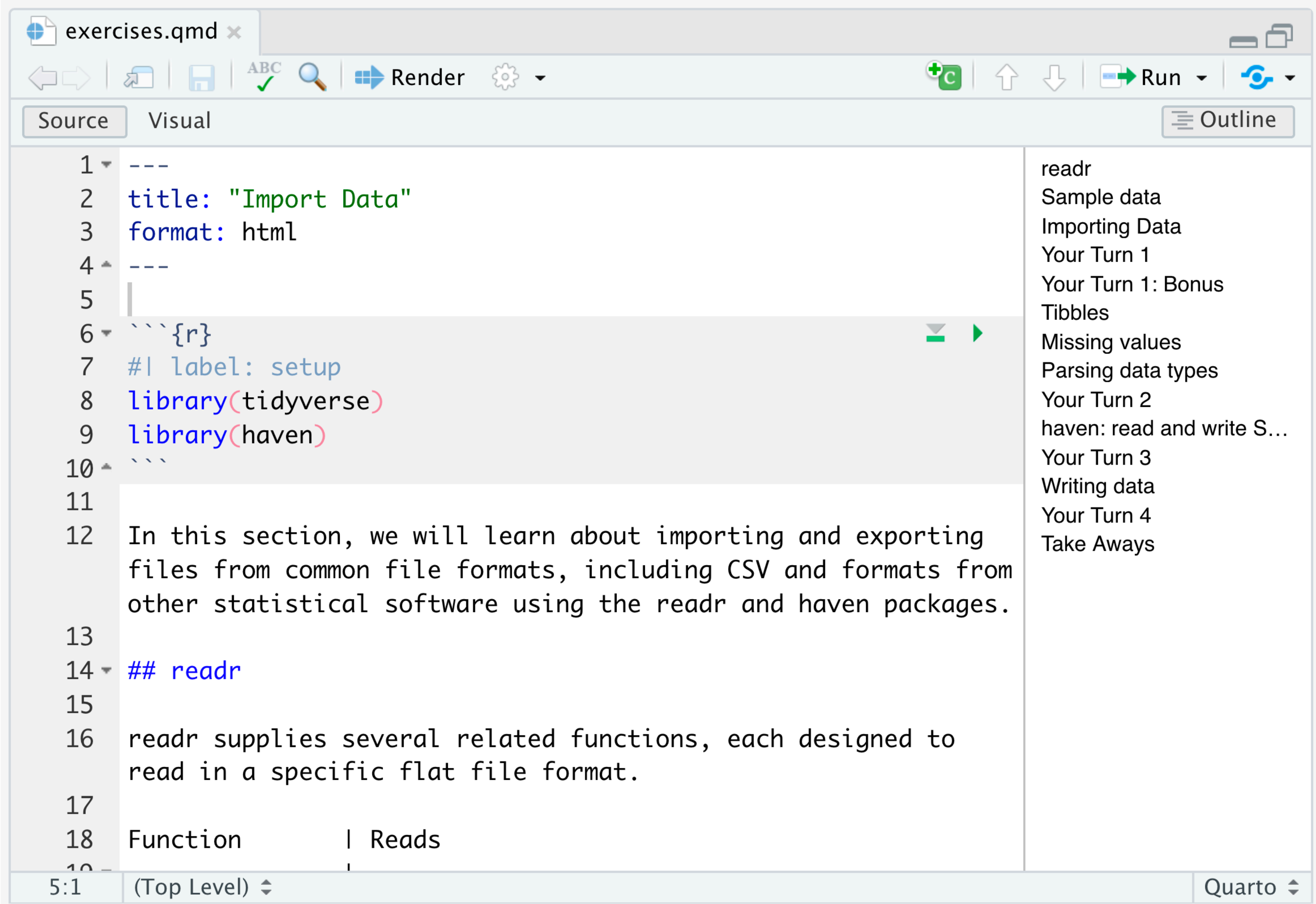
forcats, for factors.

lubridate, for dates and times.

This course is hands on!

Each section has an exercises file: *`exercises.qmd`*

exercises.qmd


exercises.qmd
Render
Run
Source
Visual
Outline
---
title: "Import Data"
format: html

```{r}
#| label: setup
library(tidyverse)
library(haven)
```
In this section, we will learn about importing and exporting files from common file formats, including CSV and formats from other statistical software using the readr and haven packages.
readr
readr supplies several related functions, each designed to read in a specific flat file format.
Function | Reads
readr
Sample data
Importing Data
Your Turn 1
Your Turn 1: Bonus
Tibbles
Missing values
Parsing data types
Your Turn 2
haven: read and write S…
Your Turn 3
Writing data
Your Turn 4
Take Aways
5:1
(Top Level)
Quarto

Code chunks

```
```{r}
csv_data <- read_csv(
  "a,b,c,d
  1,2,3,4
  5,6,7,8",
  col_types = ""
)

csv_data
```
```

Running code chunks

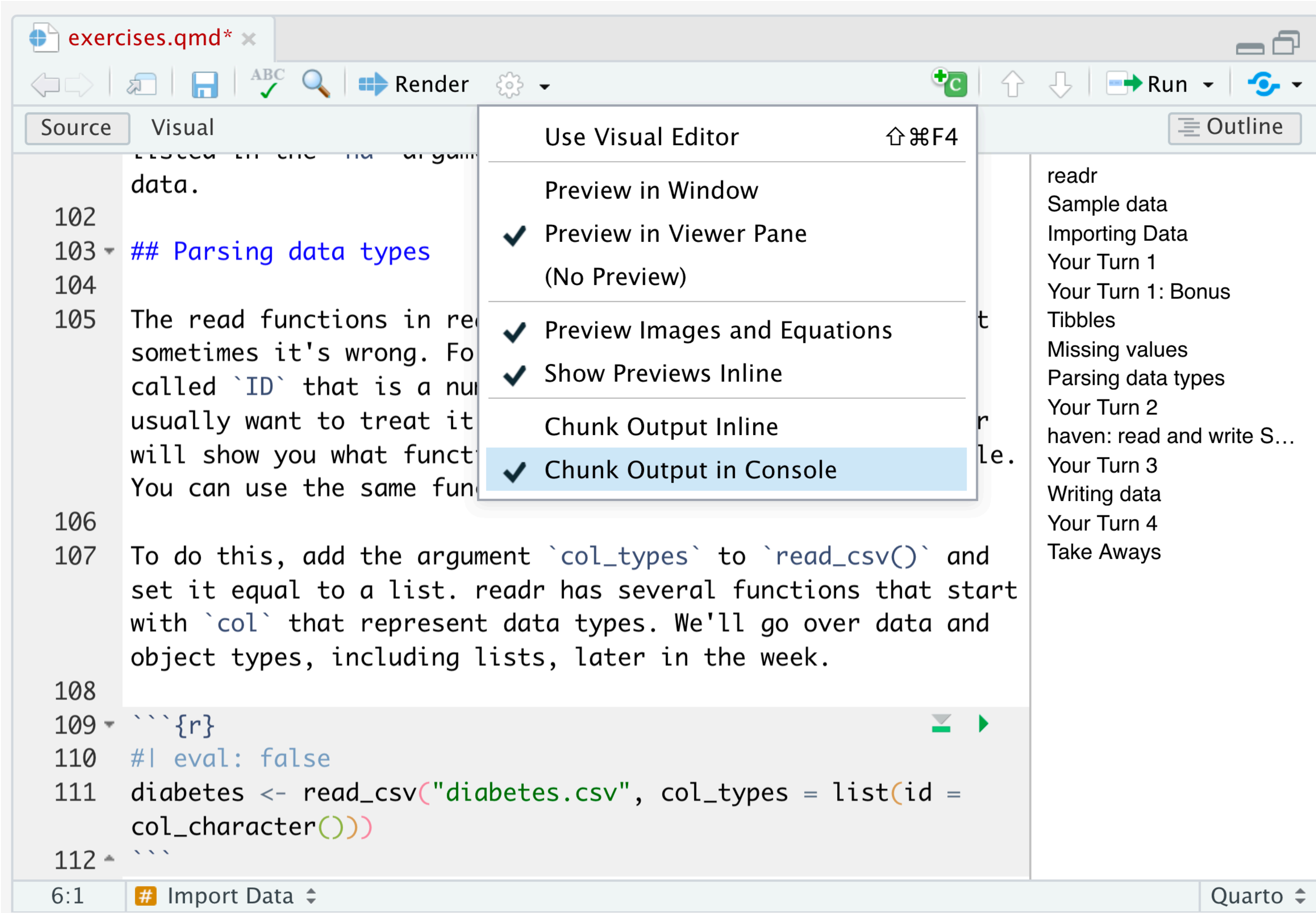
```{r}
csv_data <- read_csv(
  "a,b,c,d
  1,2,3,4
  5,6,7,8",
  col_types = ""
)

csv_data
```

a	b	c	d
<dbl>	<dbl>	<dbl>	<dbl>
1	2	3	4
5	6	7	8

2 rows

Outputting to the console



Project contents

```
├── 01-dplyr_5verbs
|   ├── cheatsheet_dplyr_5verbs.pdf
|   ├── diabetes.csv
|   ├── exercises.qmd
|   ├── slides.pdf
```